Partial Differentiation

LLE – Mathematics and Statistics Skills

Differentiation

1. For each of the functions below, $y = f(x)$, find the derived function, $\frac{\mathrm{d}y}{\mathrm{d}x} = f'(x)$

   (a) $y = 5x^3 + 10x - 8$

   (b) $y = \frac{10}{x^2} - 5\sqrt{x}$

   (c) $y = \ln 8x$

   (d) $y = e^{5x} + 3e^{x^2}$

Partial Derivatives

2. For each of the functions below, $z = f(x, y)$, find:

$$\frac{\partial z}{\partial x} \qquad \frac{\partial z}{\partial y} \qquad \frac{\partial^2 z}{\partial x^2} \qquad \frac{\partial^2 z}{\partial y^2} \qquad \frac{\partial^2 z}{\partial x\,\partial y} \qquad \frac{\partial^2 z}{\partial y\,\partial x}$$

   (a) $z = x^2 - y^2$

   (b) $z = 5x + 3x^2 - 2y^3$

   (c) $z = xy$

   (d) $z = x^2y^3 + 8x - 4$

   (e) $z = \frac{x^2}{y^2}$

   (f) $z = \ln xy$

   (g) $z = e^x + 5e^{2y}$

   (h) $z = x^2y - ye^x + xe^{-}{-}2y$

(i) $z = e^{x^2 - y^2}$

Implicit Differentiation

3. For each of the functions below, use implicit differentiation to find an expression for $\frac{\mathrm{d}y}{\mathrm{d}x}$

   (a) $x^2 + 4y = 0$

   (b) $x^2 - y^2 = 0$

   (c) $4xy + y^2 = 0$

Total Differential

4. Find the total differential, dz, of $z = f(x, y)$ using:

$$dz = \frac{\partial z}{\partial x} dx + \frac{\partial z}{\partial y} dy$$

   (a) $z = x^2 + 4y$

   (b) $z = x^2 - y^2$

   (c) $z = 4xy + y^2$

   (d) $z = \frac{4x^2}{y^3}$

5. For each of the functions in question 4, use the total derivative of $z = f(x, y)$ to find $\frac{\mathrm{d}y}{\mathrm{d}x}$ for the implicit function defined by $f(x, y) = 0$.

   Confirm that a to c match with the answers to question 3.